Law Firm Network

Cisco Packet Tracer - Step-by-Step Build Guide (Project 2)

Goal: build a secure network for a law firm: attorneys' client-confidential traffic kept in its own VLAN, IP phones on a separate voice VLAN, and an isolated Wi-Fi for clients in the conference room.

You will use: 1 Edge Router, 1 Multilayer Switch, 1 Access Switch, 1 Access Point, 5 PCs, 5 IP phones, 1 Shared Printer (plus 1 laptop to test the Guest Wi-Fi).

1. What you are building

Area	Devices	VLAN
Attorney Offices (3)	PC 1-3 + IP phone 1-3 (one PC and one phone per attorney)	10 (voice 60)
Paralegal Office	PC + IP phone 4	20 (voice 60)
Reception	PC + IP phone 5	30 (voice 60)
Shared Resources	Shared printer	40
Conference Room	Wireless Access Point (client Wi-Fi)	50
Network Cabinet	Edge Router + Multilayer Switch + Access Switch	99 (management)

Note: An IP phone has a small built-in switch port. The PC plugs into the phone and the phone plugs into the switch, so ONE switch port carries both devices: PC data in the access VLAN and phone voice in voice VLAN 60.

Key devices

Device	Hostname	Role
Access Switch	Access-SW	Connects the end devices (PCs, phones, printer, access point)
Cisco Multilayer Switch	L3-SW	Inter-VLAN routing and DHCP
Edge Router	Edge-Router	Provides the Internet connection (and the telephony service for the IP phones)

2. Network plan (cheat sheet)

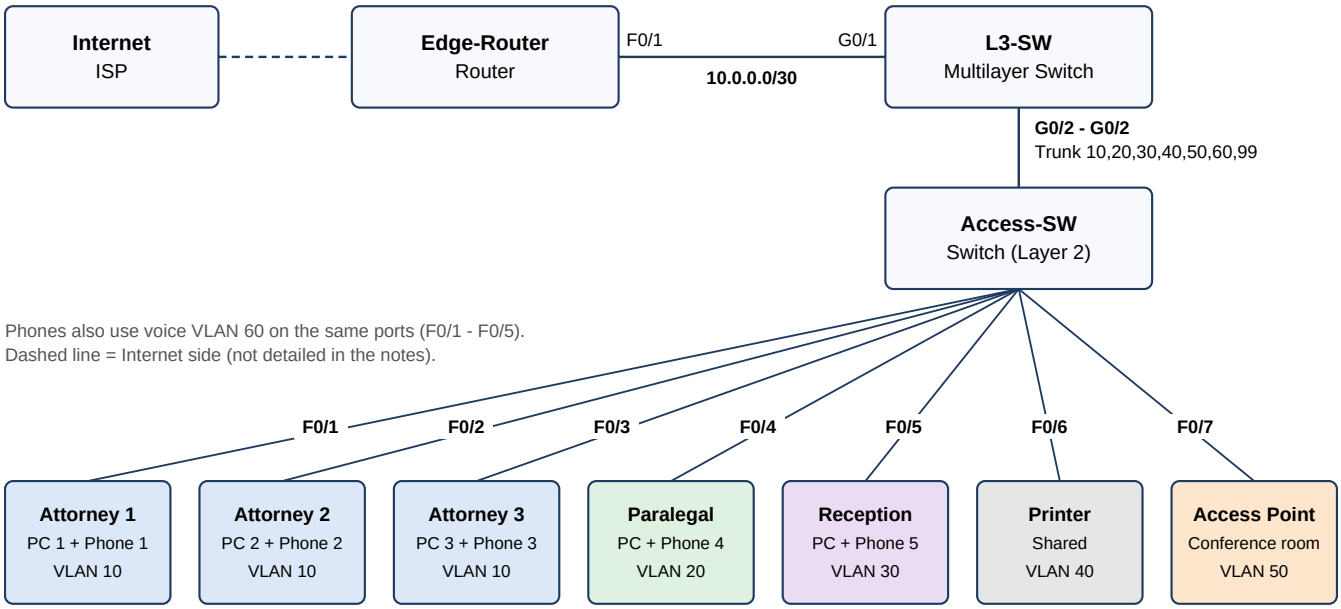
VLAN	Name	Subnet	Gateway (L3-SW)	Switch ports	Addressing
10	ATTORNEYS	192.168.10.0/24	192.168.10.1	F0/1 - F0/3	DHCP (.21+)
20	PARALEGAL	192.168.20.0/24	192.168.20.1	F0/4	DHCP (.21+)
30	RECEPTION	192.168.30.0/24	192.168.30.1	F0/5	DHCP (.21+)
40	PRINTER	192.168.40.0/24	192.168.40.1	F0/6	Static: printer = 192.168.40.10
50	GUEST-WIFI	192.168.50.0/24	192.168.50.1	F0/7	DHCP (.21+)
60	VOICE	192.168.60.0/24	192.168.60.1	F0/1 - F0/5 (phones)	DHCP (.21+); option 150
99	NETWORK MANAGEMENT	192.168.99.0/24	192.168.99.1	none (switch SVI)	Static: Access-SW = 192.168.99.2

First 20 addresses of each DHCP subnet (.1 - .20) are reserved for gateways, infrastructure devices and future static devices.

Routed link and routes

Where	Link / route	Details
Edge-Router F0/1 to L3-SW G0/1	10.0.0.0/30	Edge-Router F0/1 = 10.0.0.1; L3-SW G0/1 = 10.0.0.2
L3-SW	Default route 0.0.0.0/0	Next hop 10.0.0.1 (Edge-Router)
Edge-Router	Summary route 192.168.0.0/16	Next hop 10.0.0.2 (L3-SW)

Topology



3. Physical connections

#	From	To	Cable
1	L3 Switch G0/1	Edge Router F0/1	Ethernet
2	L3 Switch G0/2	Access Switch G0/2	Ethernet
3	IP phone 1	Access Switch F0/1	Ethernet
4	IP phone 2	Access Switch F0/2	Ethernet
5	IP phone 3	Access Switch F0/3	Ethernet
6	IP phone 4	Access Switch F0/4	Ethernet
7	IP phone 5	Access Switch F0/5	Ethernet
8	Shared Printer	Access Switch F0/6	Ethernet
9	Access Point	Access Switch F0/7	Ethernet
10	PC 1	PC port on IP phone 1	Ethernet
11	PC 2	PC port on IP phone 2	Ethernet
12	PC 3	PC port on IP phone 3	Ethernet
13	Paralegal PC	PC port on IP phone 4	Ethernet
14	Reception PC	PC port on IP phone 5	Ethernet

Note: Real life: a PoE (Power over Ethernet) capable switch gives the IP phones and the Access Point both data and electric power over the same Ethernet cable.

Tip: In Packet Tracer use copper straight-through cables between different devices and copper crossover between the two switches (or the automatic cable).

4. Multilayer switch configuration (L3-SW)

Steps 1-3: Basic setup and routing

Rename the switch, turn off DNS lookup and enable Layer 3 routing.

```
enable
configure terminal
hostname L3-SW
no ip domain-lookup
ip routing
```

Step 4: Create the VLANs

Separate attorneys, paralegal, reception, printer, guest Wi-Fi, voice and management traffic.

```
vlan 10
 name ATTORNEYS
vlan 20
 name PARALEGAL
vlan 30
 name RECEPTION
vlan 40
 name PRINTER
vlan 50
 name GUEST-WIFI
vlan 60
 name VOICE
vlan 99
 name NETWORK-MANAGEMENT
 exit
```

Step 5: Create the SVIs (default gateways)

Each Switch Virtual Interface is the default gateway for its VLAN. SVIs are shut down by default, so remember 'no shutdown'.

```
interface vlan 10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
interface vlan 20
 ip address 192.168.20.1 255.255.255.0
 no shutdown
interface vlan 30
 ip address 192.168.30.1 255.255.255.0
 no shutdown
interface vlan 40
 ip address 192.168.40.1 255.255.255.0
 no shutdown
interface vlan 50
 ip address 192.168.50.1 255.255.255.0
 no shutdown
interface vlan 60
 ip address 192.168.60.1 255.255.255.0
 no shutdown
interface vlan 99
 ip address 192.168.99.1 255.255.255.0
 no shutdown
 exit
```

Step 6: DHCP

The multilayer switch hands out IP addresses automatically. The first 20 addresses of each subnet are reserved for infrastructure devices. VLAN 40 (static printer) and VLAN 99 (management) have no pool.

```
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20
ip dhcp excluded-address 192.168.50.1 192.168.50.20
ip dhcp excluded-address 192.168.60.1 192.168.60.20
! 1st pool - ATTORNEYS
ip dhcp pool ATTORNEYS
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
! 2nd pool - PARALEGAL
ip dhcp pool PARALEGAL
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
! 3rd pool - RECEPTION
ip dhcp pool RECEPTION
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
! 4th pool - GUEST-WIFI
ip dhcp pool GUEST-WIFI
 network 192.168.50.0 255.255.255.0
 default-router 192.168.50.1
 dns-server 8.8.8.8
! 5th pool - VOICE (10.0.0.1 is the Edge-Router)
ip dhcp pool VOICE
 network 192.168.60.0 255.255.255.0
 default-router 192.168.60.1
 dns-server 8.8.8.8
 option 150 ip 10.0.0.1
 exit
```

Note: An IP phone needs to know where to get its configuration. DHCP option 150 tells the phone to get it from 10.0.0.1 (the Edge-Router), which provides the Cisco IP phone services (see section 6).

Step 7: SSH for secure remote management

SSH needs a hostname, a domain name, an RSA key and a local user. The domain name below is just an example.

```
ip domain-name lawoffice.local
username admin privilege 15 secret YOUR_PASSWORD
crypto key generate rsa modulus 1024
ip ssh version 2
line vty 0 15
 login local
 transport input ssh
 exit
```

Step 8: Trunk to the Access Switch

Interface G0/2 faces the Access-SW. One physical link carries all VLANs using VLAN tagging; the allowed list limits it to the VLANs we use (security).

```
interface g0/2
 description ## Trunk to the Access-SW ##
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,99
 no shutdown
 exit
```

> **Note:** If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line.

Step 9: Routed port to the Edge Router

Interface G0/1 faces the Edge-Router. 'no switchport' turns it into a Layer 3 routed port with an IP address.

```
interface g0/1
 description ## Routed-Link to the Edge Router ##
 no switchport
 ip address 10.0.0.2 255.255.255.252
 no shutdown
 exit
```

Step 10: Default route

If the L3 switch receives traffic for a destination that is not in its routing table, it sends it to the Edge Router.

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Step 11: Guest ACL (no access to the internal network)

The switch routes between all VLANs, so block the Guest Wi-Fi from reaching the internal networks.

```
ip access-list extended Guest-Restrictions
 deny ip 192.168.50.0 0.0.0.255 192.168.0.0 0.0.255.255
 permit ip 192.168.50.0 0.0.0.255 any
 exit
interface vlan 50
 ip access-group Guest-Restrictions in
 exit
```

ACL line	What it does
deny ... 192.168.50.0 ... 192.168.0.0 0.0.255.255	Blocks guests from every internal 192.168.x.x network
permit ip 192.168.50.0 ... any	Allows everything else = Internet only

> **Warning:** This ACL has a hidden flaw: it does not allow DHCP, so guests cannot get an IP address. Build it as written first - this is exactly the problem investigated and fixed in section 9.

Step 12: Save the configuration

So it survives a power loss.

```
end
copy running-config startup-config
```

5. Access switch configuration (Access-SW)

Steps 1-3: Basic setup and VLANs

Rename the switch, turn off DNS lookup and create the same VLANs that were created on the L3-SW.

```
enable
configure terminal
hostname Access-SW
no ip domain-lookup
vlan 10
 name ATTORNEYS
vlan 20
 name PARALEGAL
vlan 30
 name RECEPTION
vlan 40
 name PRINTER
vlan 50
 name GUEST-WIFI
vlan 60
 name VOICE
vlan 99
 name NETWORK-MANAGEMENT
 exit
```

Step 4: Management SVI and default gateway

The switch gets its own management address in VLAN 99 so it can be managed remotely.

```
interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 192.168.99.1
```

Step 5: SSH for secure remote management

Same commands as on the L3-SW.

```
ip domain-name lawoffice.local
username admin privilege 15 secret YOUR_PASSWORD
crypto key generate rsa modulus 1024
ip ssh version 2
line vty 0 15
 login local
 transport input ssh
 exit
```

Step 6: Trunk to the L3 Switch

Interface G0/2 faces the L3-SW: make it a trunk, add a description and define the allowed VLANs.

```
interface g0/2
 description ## Trunk to the L3-SW ##
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50,60,99
 no shutdown
 exit
```

> **Note:** If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line.

Step 7: Attorney ports (VLAN 10 + voice VLAN 60)

Ports F0/1 - F0/3 connect the attorneys' phones (with the PCs behind them). PortFast makes end-device ports come up fast.

```
interface range f0/1 - 3
 description ## Attorneys PHONES & PCs ##
 switchport mode access
 switchport access vlan 10
 switchport voice vlan 60
 spanning-tree portfast
 exit
```

Step 8: Paralegal, Reception, Printer and Access Point ports

Repeat the same process for the other offices. The printer and the access point have no phone, so no voice VLAN.

```
interface f0/4
 description ## Paralegal PHONE & PC ##
 switchport mode access
 switchport access vlan 20
 switchport voice vlan 60
 spanning-tree portfast
 exit
interface f0/5
 description ## Reception PHONE & PC ##
 switchport mode access
 switchport access vlan 30
 switchport voice vlan 60
 spanning-tree portfast
 exit
interface f0/6
 description ## Office Printer ##
 switchport mode access
 switchport access vlan 40
 spanning-tree portfast
 exit
interface f0/7
 description ## Guest Wireless Access-point ##
 switchport mode access
 switchport access vlan 50
 spanning-tree portfast
 exit
```

Step 9: Save the configuration

```
end
copy running-config startup-config
```

6. Edge router configuration (Edge-Router)

Steps 1-3: Basic setup and the link to the L3 Switch

Rename the router, turn off DNS lookup, then enable the interface that faces the L3-SW and give it 10.0.0.1 /30.

```
enable
configure terminal
hostname Edge-Router
no ip domain-lookup
interface f0/1
 description ## To the L3-SW ##
 ip address 10.0.0.1 255.255.255.252
 no shutdown
 exit
```

Step 4: Summary route to the internal networks

One summary route lets the Edge Router reach every internal VLAN (192.168.x.x) through the L3-SW.

```
ip route 192.168.0.0 255.255.0.0 10.0.0.2
```

Step 5: Telephony service for the IP phones

The router acts as the call controller for the Cisco IP phones. Each phone gets an extension (1001 - 1005).

```
telephony-service
 max-ephones 5
 max-dn 5
 ip source-address 10.0.0.1 port 2000
 auto assign 1 to 5
 exit
ephone-dn 1
 number 1001
 exit
ephone-dn 2
 number 1002
 exit
ephone-dn 3
 number 1003
 exit
ephone-dn 4
 number 1004
 exit
ephone-dn 5
 number 1005
 exit
```

Command	What it does
max-ephones 5	Allows up to 5 IP phones to register
max-dn 5	Allows up to 5 directory numbers (extensions)
ip source-address 10.0.0.1 port 2000	The router address (and port) the phones register to
auto assign 1 to 5	Automatically gives the phones ephone-dn 1 to 5 when they register
ephone-dn N / number 100N	Creates the extension 1001 - 1005

Step 6: Save the configuration

```
end
copy running-config startup-config
```

> **Note:** In a larger real-world deployment, Cisco voice can involve dedicated call processing platforms such as Cisco Unified Communications Manager.

> **Note:** The notes do not give the Internet-facing side of the Edge Router (ISP interface, default route, NAT). Add it only if your lab needs Internet access, using the address your ISP / cloud gives you.

7. End devices

Device	VLAN	IP address	Gateway	DNS
Shared Printer	40	192.168.40.10 (static)	192.168.40.1	8.8.8.8
PC 1 / PC 2 / PC 3	10	DHCP (automatic)	192.168.10.1	8.8.8.8
Paralegal PC	20	DHCP (automatic)	192.168.20.1	8.8.8.8
Reception PC	30	DHCP (automatic)	192.168.30.1	8.8.8.8
IP phones 1 - 5	60 (voice)	DHCP (automatic) + option 150	192.168.60.1	8.8.8.8
Guest devices (laptop)	50	DHCP (automatic)	192.168.50.1	8.8.8.8

Access Point: open its configuration and set an SSID and a pass-phrase (Wi-Fi password). Guest devices then get 192.168.50.x addresses by DHCP.

Test: try to connect the end devices (for example a laptop) to the Guest Wi-Fi.

> **Note:** The printer is in VLAN 40, which has no DHCP pool, so it gets a static IP address.

8. Quick verification

Check	Command / test
VLANs and port assignment	show vlan brief (L3-SW and Access-SW)
Trunk up, VLANs allowed	show interfaces trunk (both switches)
SVIs and routed port up	show ip interface brief (L3-SW)
Routing	show ip route (L3-SW: default route via 10.0.0.1; Edge-Router: 192.168.0.0/16 via 10.0.0.2)
DHCP pools and leases	show ip dhcp pool | show ip dhcp binding (L3-SW)
Guest ACL applied	show access-lists | show ip interface vlan 50
SSH ready	show ip ssh (both switches)
IP phones registered	show ephone (Edge-Router); phones show extensions 1001 - 1005
Attorney PC works	Ping 192.168.10.1 (gateway) and the printer 192.168.40.10
Phone call works	From one IP phone dial another extension (for example 1002)
Guest isolation works	From a guest device: ping 192.168.10.1 should FAIL

9. Troubleshooting: Guest Wi-Fi has no network access

The law firm called: devices can connect to the Guest Wi-Fi, but they are not getting any network access.

Steps 1-2: Test from the laptop

Connect a laptop to the Guest Wi-Fi and try to ping the default gateway (192.168.50.1). Then check the IP address. The laptop has no valid IP address - DHCP is not working.

```
ipconfig /all
```

Step 3: Check the Access Switch

```
enable
show vlan brief
show interface trunk
```

Step 4: Check the L3 Switch

```
enable
show vlan brief
show ip interface brief
show access-list
show ip dhcp pool
show ip dhcp binding
```

Where	Command	Look for
Access-SW	show vlan brief	VLAN 50 exists; F0/7 (Access Point) is in VLAN 50
Access-SW	show interface trunk	Trunk is up and VLAN 50 is allowed
L3-SW	show vlan brief	All VLANs exist
L3-SW	show ip interface brief	Vlan50 is up/up with 192.168.50.1
L3-SW	show access-list	Guest-Restrictions: no rule allows DHCP
L3-SW	show ip dhcp pool	GUEST-WIFI pool exists with the right network
L3-SW	show ip dhcp binding	No lease for the guest laptop

Step 5: Identify the problem

Problem found	Why it matters
The Guest-Restrictions ACL has no rule for DHCP	The ACL blocks guests from the internal network and permits them everywhere else, but it assumes the client already has an IP address. The client needs DHCP to get one - the laptop would need an IP address to send the traffic that gets it an IP address.
Guest DHCP request is dropped	A client without an IP address sends its DHCP request from 0.0.0.0 to 255.255.255.255 (UDP 68 to 67). No ACL line matches it, so it hits the implicit deny at the end of the ACL.

Step 6: Fix the ACL

Re-configure the ACL to permit DHCP (UDP 68 = DHCP client, UDP 67 = DHCP server).

```
enable
configure terminal
do show access-list
! copy the ACL name
ip access-list extended Guest-Restrictions
 permit udp any eq bootpc any eq bootps
 end
```

Tip: To be safe, place the DHCP rule first with a sequence number, for example '5 permit udp any eq bootpc any eq bootps'. Then it also matches DHCP renewals, which are sent straight to 192.168.50.1 and would otherwise hit the 'deny' line.

Step 7: Verify the fix

On the laptop, check that it now gets an address (192.168.50.x with gateway 192.168.50.1). Guest isolation must still work: ping 192.168.10.1 should fail.

```
ipconfig /all
```

Appendix: corrections made to the original notes

In the notes	Used in this guide
VLANs created with 'description ATTORNEYS' ...	name ATTORNEYS (VLANs use 'name')
'ip dhcp excluded-addresses'	ip dhcp excluded-address
'1st DHCP POOL for the ATTORNEYS' written twice	Written once
'option 150 10.0.0.1'	option 150 ip 10.0.0.1
Routed port to the Edge Router configured on 'interface g0/2' (same as the trunk)	interface g0/1 (G0/2 is the trunk to the Access-SW)
'ip access-group Guest-Restriction in'	Guest-Restrictions (must match the ACL name)
'crypto key generate rsa' then '1024' on a new line	crypto key generate rsa modulus 1024
Stray, incomplete 'ip route' line before the summary route	Removed
Password 'CISCO123'	YOUR_PASSWORD (use your own password)
'##' comments inside the DHCP configuration	'!' comments (valid in the IOS config)
'POE'	PoE (Power over Ethernet)

Added so the commands work (not written in the notes): the trunk commands on the Access-SW (the notes describe it in words), the port-by-port commands for the paralegal, reception, printer and access point ports, 'modulus 1024', 'copy running-config startup-config', the verification commands, the sequence-number tip, and the note about the Internet side of the Edge Router.